

1. Administrative & Study Identification

Full Study Title: A Multicentre, Randomised, Double-Blind, Placebo-controlled, Phase 3 Study Evaluating the Efficacy and Safety of Subcutaneous Anifrolumab in Adult Patients With Systemic Lupus Erythematosus **Short Title/Acronym:** Tulip SC
Protocol Number: D3465C00001 **NCT Number:** NCT04877691 **Sponsor Name:** AstraZeneca **Investigational Product:** Anifrolumab (Saphnelo) via Subcutaneous Injection (accessorised prefilled syringe/pen)
Phase of Development: Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of a subcutaneous treatment regimen of anifrolumab versus placebo in participants with moderately to severely active, autoantibody-positive systemic lupus erythematosus (SLE) while receiving standard of care (SOC) treatment, assessed by BICLA response at Week 52. **Secondary Objectives:** To evaluate maintained oral corticosteroid (OCS) reduction through Week 52, safety, and tolerability including injection site reactions.

2.2 Background & Rationale To address the unmet need for a more flexible, convenient, self-administered treatment option for moderate to severe SLE patients. Despite guidelines recommending earlier intervention and biologic treatments, too many people with SLE rely on oral corticosteroids, which contribute to irreversible organ damage. Subcutaneous administration offers the potential to ease patient burden compared to intravenous dosing.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target \$N\$: 367 participants
Number of Sites: Multinational (United States + 14 others)
Estimated Study Start: 08-Jun-2021 **Estimated Primary Completion:** 22-Aug-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 to ≤ 70 years. 2. Patients who have a diagnosis of adult SLE according to the ACR

1997 revised criteria for ≥ 24 weeks prior to signing the ICF. 3. To be eligible a patient must have SLEDAI-2K ≥ 6 points and "Clinical" SLEDAI-2K score ≥ 4 points at screening, and BILAG2004 level A disease in ≥ 1 organ system or level B disease in ≥ 2 organ systems.

4.2 Exclusion Criteria 1. Active severe or unstable neuropsychiatric SLE. 2. Active severe SLE-driven renal disease. 3. History of any non-SLE disease that has required treatment with oral or parenteral corticosteroids for more than a total of 2 weeks within the last 24 weeks prior to signing the ICF.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Fixed subcutaneous dose of anifrolumab or placebo administered once weekly. **Route of Administration:** Subcutaneous (SC) via an accessorised prefilled syringe (aPFS) or pre-filled pen. **Duration of Treatment:** 52 weeks for the double-blind period, with an optional 52-week open-label extension (OLE).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care (SOC) treatment, consisting of 1 or any combination of the following: oral glucocorticoids, antimalarials, and/or immunosuppressants. **Prohibited:** Use of restricted medications beyond the protocol-allowed threshold before assessment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, SLE Disease Activity Assessments (SLEDAI-2K, BILAG2004), Labs
Baseline	Day 1 (Week 0)	Randomization, First Dose, Initiation of SOC maintenance
Interim	Week 40 $\pm$ 7	Safety Labs, Efficacy Assessment, Evaluation of OCS reduction target (≤ 7.5 mg/day)
End of Study	Week 52	Final primary endpoint evaluation (BICLA response), Optional transition to Open-Label Extension (OLE)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: British Isles Lupus Assessment Group-based Composite Lupus Assessment (BICLA) response at Week 52. **Measurement Method:** Composite binary endpoint requiring reduction of baseline BILAG-2004 A to B/C/D and baseline BILAG-2004 B to C/D, with no worsening in other organ systems.

7.2 Secondary & Exploratory Endpoints 1. Maintained oral corticosteroid (OCS) reduction among patients with baseline OCS ≥ 10 mg/day (achieving ≤ 7.5 mg/day by Week 40 and

maintaining to Week 52). 2. Incidence of adverse events (AEs) including injection site reactions and severe infections.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of adverse events, particularly injection site reactions, opportunistic infections, and herpes zoster. **Serious Adverse Events (SAEs):** Standard protocol timeline reporting (typically 24 hours); tracking any infections requiring hospitalization or IV antimicrobial treatment. **Data Monitoring Committee (DMC):** Internal safety monitoring managed by AstraZeneca to oversee the risk profile in comparison to IV formulation.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy (approx. 367 randomized participants). **Sample Size Justification:** Powered to detect a statistically significant difference in BICLA response between SC anifrolumab and placebo. **Handling of Missing Data:** Standard imputation rules applied for patients discontinuing investigational product or using restricted medications beyond the allowed threshold, generally treated as non-responders.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular clinical monitoring of investigator sites to ensure protocol adherence, especially regarding background SOC tapering regimens and clinical scoring scale accuracy (BILAG/SLEDAI). **Audit Trail:** Data entered into validated eCRF systems with continuous data clarification tracking.

11. Study Identifiers & Governance

Primary ID: D3465C00001 **Registry Number:** NCT04877691 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** Tulip SC **Official Regulatory Title:** A Multicentre, Randomised, Double-Blind, Placebo-controlled, Phase 3 Study Evaluating the Efficacy and Safety of Subcutaneous Anifrolumab in Adult Patients With Systemic Lupus Erythematosus

12. Clinical Framework (SLE Specific)

Primary Condition: Systemic Lupus Erythematosus (SLE) **Diagnosis Threshold:** SLEDAI-2K ≥ 6 points and BILAG2004 level A disease in ≥ 1 organ system or level B disease in ≥ 2 organ systems **Medical Methodology:** Biologic addition to Standard of Care (oral glucocorticoids, antimalarial, and/or immunosuppressants) **Optimization Target:** BICLA response and Maintained Oral Corticosteroid (OCS) reduction

13. Study Procedures & Methodology

Management Pathway: Weekly Subcutaneous self-administration of anifrolumab **Education Protocol (HCP):** Training on composite SLE clinical scoring (BILAG2004, SLEDAI-2K, PGA) **Education Protocol (Patient):** Self-learning and onsite training for safe use of accessorised prefilled syringe (aPFS) / pre-filled pen at home **Quality Audit Mechanism:** Adherence to weekly injection tracking and monitoring for injection site reactions (ISR)

14. Observation & Follow-Up Schedule

Total Duration: 52 weeks (double-blind) + 52 weeks (Open-Label Extension) **Onsite Visit Cadence:** Routine clinic visits up to Week 52 for clinical efficacy scoring and steroid tapering assessment **Remote Monitoring Frequency:** Standard safety check-ins and compliance tracking for at-home SC injections **Checklist Review Interval:** Every protocol-defined visit block (e.g., Week 40 baseline check for OCS reduction target)

15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	Susan Manzi, MD, MPH	_____	04-Apr-2026
Clinical Operations Lead	AstraZeneca Clinical Operations Lead	_____	04-Apr-2026
Lead Statistician	AstraZeneca Lead Statistician	_____	04-Apr-2026